Q1	A B **C** D E F
Q2	A B C D **E** F
Q3	A **B** C D E F
Q4	A B **C** D E F
Q5	A B C **D** E F
Q6	A B C D **E** F
Q7	A B C D E **F**
Q8	A B C D **E** F
Q9	A B C D **E** F
Q10	A **B** C D E F

1. What is the correct scientific formula used to calculate "Work Done"? [1]

Work = Speed ×Time

Work = Mass × Gravity

Work = Force × Distance

Work = Force + Distance

Work = Distance − Force

Work = Force ÷ Distance

2. According to the text, why might a bicycle use a larger gear when riding? [1]

To increase the top speed for racing.

To stop the bike quickly.

To make the bike lighter.

To prevent the chain from falling off.

To increase the force to help climb hills.

To reduce the friction on the tires.

3. Wheels are useful because they reduce which force that normally resists motion? [1]

Magnetism

Friction

Electricity

Thrust

Gravity

Tension

4. A screw is essentially which other simple machine wrapped around a cylinder? [1]

A wedge

A lever

An inclined plane

A wheel and axle

A pulley

A gear

5. In a lever system, what do we call the fixed point that the lever turns or rests on? [1]

The beam

The handle

The force

The fulcrum

The load

The plank

6. Which statement is true about using an inclined plane (ramp) with a shallow slope compared to a steep slope? [1]

It reduces the distance the object travels.

It eliminates gravity completely.

It requires the same amount of force as lifting it straight up.

It increases the weight of the object.

It requires less force, but the object must travel a longer distance.

It requires more force to move the object.

7. A wedge is designed to focus force into a smaller area. What is a common use for a wedge? [1]

To measure time.

To connect two gears together.

To lift an elevator.

To turn a doorknob.

To roll a car down a hill.

To split an object apart.

8. If Gear A turns in a clockwise direction, in which direction will the connected Gear B turn? [1]

Upwards

Downwards

It will not move

Clockwise

Anti-clockwise

Linearly

9. What is the main advantage of using a system of multiple pulleys to lift a load? [1]

It increases the gravity acting on the object.

It makes the object heavier.

It increases the speed of the lift.

It makes the rope shorter.

It reduces the force needed to lift the load.

It stops the object from moving.

10. How does a doorknob (wheel and axle) make it easier to open a door? [1]

By removing the need for an axle.

By allowing you to apply a small force over a large distance to turn the axle.

By using a counterweight.

By increasing the friction on the lock.

By acting as a wedge to split the door open.

By reducing the size of the door.